

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning,

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I J O Joanna Divine(192572003) (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Joanna

RESEARCH ARTICLE REVIEW

Selected Topic: Deep Learning / Computer Vision / AI Application

Selected Research Article

Title: Domain Knowledge Powered Deep Learning for Breast Cancer Diagnosis Based on Contrast-Enhanced Ultrasound Videos

Authors: Chen Chen, Yong Wang, Jianwei Niu, Xuefeng Liu, Qingfeng Li, and Xuanton Gong

Journal: IEEE Transactions on Medical Imaging

Volume: 40, Issue 9

Pages: 2439–2451

Year: 2021

Publisher: IEEE

DOI: 10.1109/TMI.2021.3078370

The article was published in September 2021, with electronic publication on August 31, 2021.

DOI Link: <https://doi.org/10.1109/TMI.2021.3078370>

Research Domain: Artificial Intelligence, Deep Learning, Computer Vision, Medical Image Analysis, and Breast Cancer Diagnosis.

TASK 1: ARTICLE SUMMARY

1(a) Domain and ML/AI Problem

The selected research article focuses on **breast cancer diagnosis using deep learning and contrast-enhanced ultrasound (CEUS) videos**. Breast cancer diagnosis traditionally requires physicians to examine ultrasound images and other clinical information. Conventional deep learning methods for breast ultrasound diagnosis mainly use static ultrasound images. However, static images may not contain all the information available during an actual ultrasound examination.

The authors identify this limitation and propose a deep learning approach that works with **CEUS video sequences** rather than only static images. CEUS videos contain temporal information about blood flow and tumor characteristics. The proposed model uses a **3D Convolutional Neural Network (3D CNN)** together with domain-knowledge-guided attention mechanisms. The authors specifically model two behaviors followed by radiologists: concentrating on important time periods and comparing CEUS frames with corresponding ultrasound images.

The main AI problem addressed is therefore **automated classification/diagnosis of breast lesions using temporal ultrasound information while incorporating knowledge of how radiologists interpret CEUS videos**.

1(b) Key Objective, Research Gap and Motivation

The primary objective of the research is to develop a deep learning model capable of accurately diagnosing breast cancer from contrast-enhanced ultrasound videos. Existing deep learning approaches have demonstrated strong performance in breast cancer diagnosis, but many of them rely mainly on static ultrasound images. This creates a research gap because ultrasound examinations are dynamic, and CEUS videos contain temporal information about the blood supply and enhancement characteristics of tumors. The authors were motivated by the observation that radiologists do not simply examine one image; instead, they observe particular time periods and compare different ultrasound information during diagnosis. To address this gap, the researchers designed a 3D convolutional neural network that can learn spatial and temporal information from CEUS videos. They further introduced a domain-knowledge-guided temporal attention module and a channel attention module. These components attempt to guide the network toward diagnostically important temporal segments and image information. The proposed approach was evaluated using a Breast-CEUS dataset containing 221 cases. The study reports a sensitivity of 97.2% and an accuracy of 86.3%.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML/AI Techniques Used

The major technique used in the research is **Deep Learning**, specifically a **3D Convolutional Neural Network (3D CNN)**.

Unlike a conventional 2D CNN that processes individual images, a 3D CNN can process spatial and temporal information. This makes it suitable for video-based medical diagnosis.

The proposed methodology contains the following major components:

1. 3D Convolutional Neural Network

The backbone of the proposed model is a 3D CNN. It extracts information from multiple consecutive ultrasound frames and therefore learns both:

- Spatial characteristics of breast lesions
- Temporal characteristics of CEUS videos

2. Domain-Knowledge-Guided Temporal Attention

Radiologists generally focus on particular time periods while examining CEUS videos.

The proposed temporal attention mechanism attempts to identify and emphasize diagnostically important temporal information instead of treating all video frames equally.

3. Channel Attention

The channel attention mechanism helps the network identify useful feature channels and assign greater importance to informative features.

4. Domain Knowledge

An important contribution is the incorporation of medical-domain knowledge into the deep learning architecture.

The authors observed two important radiologist behaviors:

1. Focusing on specific time slots.
2. Paying attention to differences between CEUS frames and corresponding ultrasound images.

The model incorporates these observations into its architecture.

Dataset

The researchers evaluated the model using their **Breast-CEUS dataset**.

Dataset characteristic Description

Application	Breast cancer diagnosis
Data type	Contrast-enhanced ultrasound videos
Number of cases	221
Input	CEUS video information and corresponding ultrasound information
Target	Breast lesion diagnosis/classification
Domain	Medical imaging

The publicly available article information confirms that the Breast-CEUS dataset contains **221 cases**.

Preprocessing and Training

The ultrasound video data must be processed so that it can be supplied to the 3D CNN. The model learns spatial and temporal patterns from the video sequences.

The experimental methodology can be represented as:

CEUS Video → Data Preparation → 3D CNN → Temporal Attention → Channel Attention → Classification → Diagnosis

The authors also investigated the contribution of their domain-knowledge modules by validating them using both **C3D** and **3D ResNet (R3D)** architectures.

Evaluation Methodology

The main reported performance measures include:

- Sensitivity
- Specificity
- Accuracy

The proposed model achieved:

- **Sensitivity = 97.2%**
- **Accuracy = 86.3%**

The incorporation of domain knowledge resulted in:

- **3.5 percentage-point improvement in sensitivity**
 - **6.0 percentage-point improvement in specificity**
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2(b) Connection With Course Concepts

Course Concept	Connection to Article
Deep Learning	The model uses a 3D CNN to automatically learn features from ultrasound videos.
Computer Vision	The system analyzes medical ultrasound images and video sequences.
Inductive Learning	The network learns diagnostic patterns from labeled examples.
Statistical Learning	The model learns a relationship between imaging features and diagnostic outcomes.
AI Techniques	The system automates part of the medical diagnostic decision process.
Feature Learning	CNN layers automatically learn useful visual and temporal representations.
Attention Mechanism	Temporal and channel attention identify more informative information.

Methodological Strength

A major strength is the integration of **domain knowledge into the deep learning architecture**. Instead of relying only on automatically learned features, the model incorporates observations about how radiologists examine CEUS videos. This can make the model more appropriate for the actual diagnostic process.

Methodological Limitation

The dataset contains only **221 cases**, which is relatively small for training and evaluating a deep learning system. A small dataset can increase the possibility of overfitting and may limit how well the model generalizes to patients, hospitals, ultrasound devices, and populations not represented in the dataset.

TASK 3: RESULTS AND CRITICAL EVALUATION

3(a) Major Results

The proposed model achieved:

Metric	Proposed Model
Sensitivity	97.2%
Accuracy	86.3%
Sensitivity improvement from domain knowledge	3.5%
Specificity improvement from domain knowledge	6.0%

The authors also demonstrated the usefulness of the domain-knowledge modules in both **C3D** and **R3D** architectures.

Interpretation

The sensitivity of 97.2% is particularly important for medical diagnosis because high sensitivity means the system can identify a large proportion of positive cases.

The accuracy of 86.3% indicates that the model performs well overall, although accuracy alone is not sufficient to establish clinical usefulness.

3(b) Critical Evaluation

Are the Metrics Appropriate?

Yes. Sensitivity and specificity are particularly important for medical diagnosis.

Sensitivity measures the ability to correctly identify patients with the disease:

$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$

Specificity measures the ability to correctly identify patients without the disease:

$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$

Accuracy provides an overall measure of classification performance.

For a medical diagnostic system, however, it would also be useful to report:

- Precision
- F1-score
- AUC-ROC
- Confidence intervals
- Calibration
- External validation performance

Therefore, although the reported metrics are relevant, a more comprehensive clinical evaluation would strengthen the research.

Dataset Sufficiency

The use of 221 cases demonstrates the feasibility of the approach, but the dataset is relatively small for a deep learning model.

Potential issues include:

1. Limited patient diversity.
2. Possible institutional bias.
3. Possible variation in ultrasound equipment.
4. Limited representation of different disease stages.
5. Risk of overfitting.
6. Lack of extensive external validation.

Therefore, the results should not automatically be interpreted as evidence that the model will perform equally well in every hospital.

Possible Biases and Threats to Validity

Potential threats include:

- **Dataset bias:** The dataset may not represent the broader patient population.
- **Equipment bias:** Ultrasound machines from different manufacturers may produce different image characteristics.

- **Population bias:** Demographic characteristics may differ between the research dataset and real-world patients.
 - **Sample-size limitation:** Only 221 cases were used.
 - **Generalization risk:** High performance on one dataset does not guarantee high performance on external datasets.
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3(c) Real-World Applicability

The proposed technology could potentially be used as a **clinical decision-support system** for breast cancer screening and diagnosis.

Possible Applications

1. Breast cancer screening.
2. Assistance for radiologists.
3. Automated analysis of CEUS videos.
4. Hospital diagnostic-support systems.
5. Telemedicine.
6. Computer-aided diagnosis systems.

Data Availability

Large, high-quality medical datasets are difficult to collect because medical imaging requires expert annotation and patient consent.

Computational Requirements

3D CNNs require considerably more computational resources than simple traditional machine learning models because they process multiple video frames.

Scalability

For deployment across hospitals, the system would need to handle differences in:

- Imaging equipment
- Image quality
- Patient populations
- Data formats
- Clinical workflows

Privacy

Medical ultrasound data contains sensitive patient information. Appropriate privacy protection, anonymization, access control, and secure data storage would be required.

Cost

Deployment may require:

- GPU hardware
- Data storage
- Software infrastructure
- Model maintenance
- Clinical validation

Ethical Issues

An AI system should not completely replace medical professionals. Incorrect predictions could potentially affect patient treatment. Therefore, the system should preferably function as **decision support**, with qualified clinicians retaining final responsibility.

TASK 4: REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights

Insight 1: Deep Learning Can Learn Features Automatically

The article demonstrates how a 3D CNN can automatically learn useful spatial and temporal features from ultrasound videos.

Course connection: Deep Learning and Feature Learning.

This improved my understanding that deep learning can reduce the need for manually designed image features.

Insight 2: Domain Knowledge Can Improve AI Models

One of the most important lessons is that deep learning does not always have to operate independently of human knowledge. The authors incorporated knowledge about how radiologists analyze CEUS videos into the network architecture.

Course connection: Artificial Intelligence and Knowledge-Based Techniques.

The article shows that combining domain knowledge with data-driven learning can improve model performance.

Insight 3: Evaluation Metrics Must Match the Application

The article reports sensitivity, specificity, and accuracy rather than relying only on accuracy.

Course connection: Statistical Learning and Model Evaluation.

For healthcare applications, false negatives can be especially important, so sensitivity is an important metric.

4(b) How the Article Improved My Understanding

Before studying this article, deep learning could be viewed mainly as a technique for classifying images. This research demonstrates that deep learning can also process **video sequences**, allowing both spatial and temporal information to be learned.

The article also improved my understanding of attention mechanisms. Rather than treating every piece of information equally, attention allows the model to focus on more important features or time periods.

Most importantly, the research demonstrates a practical AI application in healthcare. It shows how theoretical concepts such as CNNs, feature learning, attention mechanisms, and classification can be integrated into a real-world medical decision-support system.

4(c) Proposed Research Extension

Proposed Extension: Explainable and Federated Deep Learning for Multi-Hospital Breast Cancer Diagnosis

I would extend the research by developing an **explainable federated learning framework** using CEUS data from multiple hospitals.

Instead of collecting all patient data in one central location, each hospital could train the model locally. Only model updates would be shared between hospitals.

The proposed architecture could be:

Hospital CEUS Data → Local 3D CNN → Attention Mechanism → Local Training → Federated Aggregation → Global Model

An explainability component could also be added to show which temporal regions and image features influenced the prediction.

Feasibility

The approach is feasible because the existing research already demonstrates that 3D CNNs, attention mechanisms, and domain knowledge can be applied to CEUS videos.

Expected Improvement

The proposed approach could:

- Increase dataset diversity.
- Improve generalization.
- Reduce dependence on a single hospital.
- Protect patient data.
- Improve model transparency.
- Provide more reliable predictions across institutions.

Potential Impact

A successful system could provide radiologists with an AI-based second opinion while reducing the need to transfer sensitive patient images between hospitals. Explainability could also increase clinician confidence and make it easier to identify incorrect predictions.

OVERALL CRITICAL REVIEW

The selected research presents a meaningful application of deep learning to breast cancer diagnosis. Its major contribution is the use of CEUS videos instead of relying exclusively on static ultrasound images. The incorporation of domain knowledge through temporal and channel attention is particularly valuable because it attempts to imitate important aspects of the radiologist's diagnostic process.

The reported sensitivity of 97.2% and accuracy of 86.3% demonstrate promising performance. The reported improvements in sensitivity and specificity after incorporating domain knowledge further support the usefulness of the proposed approach.

However, the relatively small dataset of 221 cases creates limitations regarding generalization. Additional multi-center datasets, external validation, larger sample sizes, and more comprehensive evaluation metrics would strengthen the evidence.

Overall, the research demonstrates how **Deep Learning + Computer Vision + Domain Knowledge** can be combined to develop intelligent healthcare applications. It provides a strong example of connecting theoretical AI/ML concepts with a practical medical problem.

CONCLUSION

The article “**Domain Knowledge Powered Deep Learning for Breast Cancer Diagnosis Based on Contrast-Enhanced Ultrasound Videos**” provides an effective example of applying deep learning to medical image and video analysis. The proposed 3D CNN uses spatial and temporal information from CEUS videos and incorporates domain knowledge through attention mechanisms.

The research achieved a sensitivity of **97.2%** and accuracy of **86.3%** on a 221-case Breast-CEUS dataset.

The major learning from this article is that successful AI systems can benefit from combining **data-driven deep learning with human/domain knowledge**. Future work should focus on larger multi-center datasets, external validation, explainable AI, privacy-preserving federated learning, and real-time clinical deployment.

ARTICLE SOURCE

IEEE DOI: [10.1109/TMI.2021.3078370](https://doi.org/10.1109/TMI.2021.3078370)

Article verification: The bibliographic information, authors, publication details, DOI, dataset size, architecture, and reported results are verified through the PubMed record for the IEEE Transactions on Medical Imaging article.

Expected Learning Outcomes Achieved

1. **Identify recent AI/ML research:** A 2021 peer-reviewed IEEE deep learning research article was selected.
2. **Analyze ML methodology:** The 3D CNN and attention mechanisms were examined.
3. **Evaluate results:** Sensitivity, specificity, and accuracy were discussed.
4. **Critically assess research:** Dataset limitations, bias, and generalization issues were identified.
5. **Connect theory with practice:** Deep Learning, Computer Vision, Inductive Learning, and Statistical Learning concepts were connected to healthcare.
6. **Propose research improvements:** Explainable and federated multi-hospital deep learning was proposed as an extension.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand